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A REVIEW OF THE FERROELECTRIC MODEL OF MICROTUBULES

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It has been 25 years since the piezoelectric properties of the microtubule were first discussed. Subsequent to the discovery of its piezoelectric properties, its consequences have been observed in experiments which have shown microtubules to align under the influence of electric fields. In this regard, our investigations have sought to determine in which situations the ferroelectric properties are important for the function of the microtubule, and in addition, how the microtubule may be affected by the transient endogenous electric fields of living cells. We report a summary of recent findings and discuss their implications.

1. INTRODUCTION

Microtubules have recently generated much interest well beyond the biological community^[1–3] because they demonstrate how physical interactions are important within living systems. The microtubule is a polymer of a protein known as tubulin which is common to most animal, plant and protist cells. Microtubules are one of three main structures which form the cytoskeleton that supports the cell. Microtubules perform many functions related to both their structural strength and flexibility. Other than serving as a cellular backbone, they act as a railroad for the transportation of cellular materials. For this purpose the orientation of the microtubule is essential since the motor proteins which carry materials toward or away from the cell body may travel in only one direction along a microtubule. Hence, the permanent dipole moment of the tubulin molecules may be important in the guidance of molecules, known as motor proteins, that *walk* along the microtubule surface. More recently, microtubules have been implicated in

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simple signalling within the cell. In protists, which are single-cell organisms, information processing cannot occur through any sort of multicellular neural networking, microtubules have been proposed as the most likely organelle to accomplish this task. These curious possibilities are explained in this review from the standpoint of ferroelectrics.

The piezoelectric properties of the microtubule were first discovered by Athenstädt, who was searching for piezoelectric effects in living systems.^[4] The measurement of its piezoelectric properties and subsequent classification as a piezoelectric polymer declared the existence of a permanent electric dipole on the microtubule. Since the microtubule is composed of many identical subunits (α - β tubulin dimers), each much itself possess a permanent dipole. This has been confirmed by the experiments of Vassilev *et al.*^[5] and subsequently by Kaimanovich *et al.*^[6, 7] and White *et al.*^[8] which have shown that arrays of microtubules may be oriented by electric fields and have gone on to speculate that intracellular electric fields may induce microtubule orientation. Further experiments have demonstrated that the tubulin dimer of which the microtubule is composed has two different conformational states within the microtubule.^[9] When guanosine triphosphate (GTP) hydrolysis occurs, the length of the tubulin dimer is reduced from 8.3 nm to 8.0 nm. With any atomic reorganization such as that involved in a conformational change of a molecule, the electrons must also be redistributed. Given that the surface of tubulin is strongly negatively charged at cellular pH and that its size changes by a significant amount on the scale of electronic interactions, this structural compression must result in a change of the dipole moment. This electron redistribution has also spurred interest in charge conduction along microtubules.^[10] However, the length scale of tubulin implies that microtubules lie in the gray area between quantum and classical physics. To the best of our knowledge, the first theoretical model of ferroelectricity in microtubules has been proposed by Satarić *et al.*^[11] and it was based on an effective field approximation in the continuum limit giving rise to travelling ferroelectric domain walls.

Our present model of ferroelectric phases in microtubules is simply a classical Ising model where: (a) the interaction constants are determined by classical dipole-dipole electromagnetic forces; (b) the 2D lattice is periodic in one direction so as to form a tube; and (c) the dipole moment has only two possible orientations, though they are not precisely opposite to each other. The last property shall be important when interactions between the dipole of tubulin and transient endogenous electric fields are considered; the dipole moment must have a radial component so that the dipole interaction with the external field is non-zero.

Some of the interest in microtubule ferroelectric properties has been sparked by their unusual assembly properties, their special arrangement in cells of various types, their involvement with cell division and their suspected role in some cellular signalling. We elaborate upon these examples of microtubule structure and function.

- *In vitro*, microtubules experience a form of stochastic assembly known as dynamic instability.^[12] Typically, individual microtubules are either undergoing slow steady growth or extremely rapid shortening, known as a catastrophe event. The entire pool of tubulin in the cell, however, exhibits smooth oscillations in the quantity of tubulin which is assembled. Electromagnetic interactions resulting from dipoles on assembled microtubules and on free tubulin dimers have been proposed as one coupling mechanism to describe the transition from individual to ensemble dynamics. In addition, the catastrophe event may be explained. The idea is that the dipole interactions induce a conformation with some *fault* energy. At some point, the fault energy is sufficient to induce a catastrophe event.
- The arrangement of microtubules in cells is always such that the negative end is closest to the center of the cell and the positive end extends toward the cell membrane, or down the long extension of a nerve cell known as an axon.^[13] The positive end of the microtubule should not be confused as denoting the direction in which a microtubule's dipole is pointing since at this point the direction has not been experimentally determined. However, this convention is consistent with the biological literature on the subject and the plus end indicates the orientation of the structure. Perhaps 5% or fewer microtubules will typically be improperly oriented. Given that cell membranes have a resting potential caused by ionic gradients across the membrane, it is unlikely that the electrical potential drops precisely to zero immediately inside the membrane. There is a small residual electric field on the order of 1–10 V/m, which one suspects aids in the efficient alignment of microtubule orientation within the cell. This alignment is crucial to motor proteins which travel along microtubules and specifically carry material toward or away from the centrosome which is located close to the cell nucleus.
- In cells which are undergoing division, microtubules align themselves from the two poles of the cells and extend toward the metaphase plate which will ultimately divide the two daughter cells. Pairs of chromosomes are held together along the plate with two microtubules, one anchored near each pole. The pattern of microtubules is highly reminiscent of the

field lines between a positive and negative charge. This has reinforced the speculation that electromagnetic effects are at work, given that it is known that microtubules may be ordered by electric and magnetic fields^[14] and has driven investigators to study charge flow in the cell membrane before and during cell division. More intriguing than the array of microtubules which forms is the ability of the microtubules to synchronize chromosome separation and subsequently initiate the segregation process as the microtubules themselves disassemble. It seems necessary for the microtubules to interpret or carry some signal in this process since a signal carried by a chemical messenger, through diffusion, could not arrive at the chromosomes with the perfect synchrony that is observed. If the signal is mechanical, then the structure which supplies the push or pull on the microtubules must be found, an alternative explanation in terms of piezoelectric or ferroelectric coupling seems more reasonable.

Finally, it has recently been shown that microtubules and associated members of the cytoskeleton play a role in intracellular signalling.^[15, 16] The mechanism of signal transduction is not understood at present and may again be intimately associated with the ferroelectric properties of the polymer.

2. THE MICROTUBULE LATTICE

The lattice of the microtubule is helical. Within cells, it is most commonly composed of 13 columns, known as protofilaments. The outer ring of protofilaments is 25 nm in diameter, the inner ring 15 nm (Fig. 1). The interior of a microtubule is hollow and they may have an arbitrary length; however, microtubules with a length of a few microns are not uncommon. Since the length of each dimer is about 8 nm, the typical microtubule has perhaps 100–1000 rows. Each α - β tubulin dimer is composed of two nearly identical monomers, individually known as α and β tubulin. This lattice is arranged such that there is an upward offset between neighbouring protofilaments of 0.92 nm (12 nm/13) when moving around the microtubule in a left-handed sense. Since the monomer spacing along a protofilament is 4 nm, this represents a rise of 3 monomers for each loop of the helix. However, there are two types of lattice which have been described in the literature, known as the A lattice and the B lattice. If one follows the helix with the small 0.92 nm offset (known as the $\bar{3}$ start helix), the monomers alternate (α - β - α - β -...) in the A lattice but remain the same in the B lattice (α - α - α - α -...), except at the seam, where there is an α - β link.

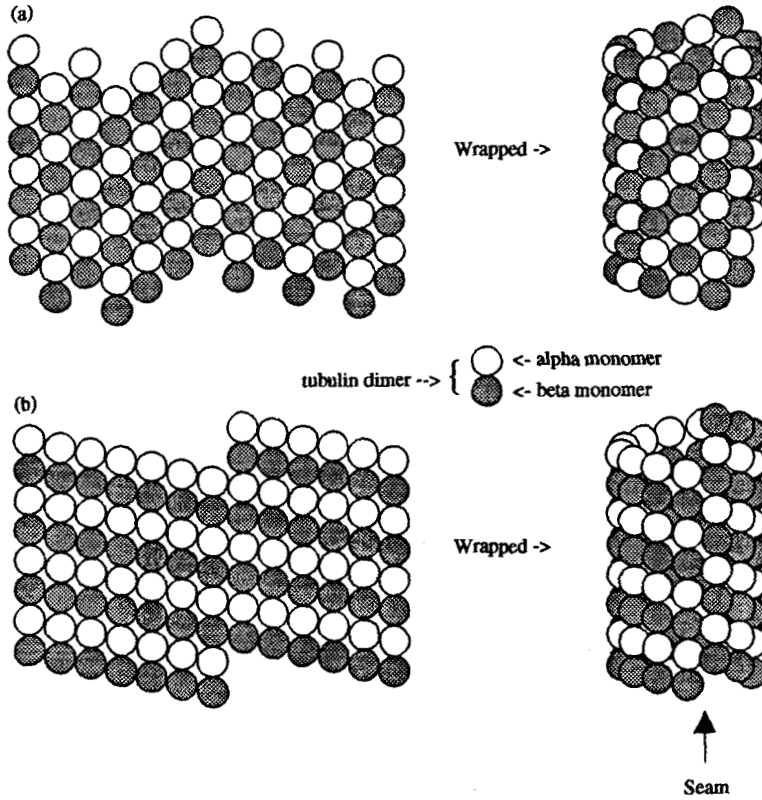


FIGURE 1 The two possible microtubule lattices are depicted. Typical microtubules are much longer than these which are shown with a length of only 4 dimers along each of the 13 protofilaments (columns). (a) The A lattice, which has helical symmetry. (b) The B lattice, which has a seam where a line following the α -monomers around the microtubule meets a β -monomer. The left side depicts unwrapped (non-physical) microtubules; the figures resulting from our simulations are given in a similar manner.

Thus the A lattice has complete helical symmetry which the B lattice does not possess. So, the monomer lattices are identical but the dimer lattices are quite different. Since the dimer is the fundamental unit from which microtubules are constructed, the dipole is associated with each dimer and the lattice change means that interactions are quite different between the two geometries.

Our model of ferroelectric effects in microtubules was inspired by the cellular automaton model of Hameroff *et al.*^[2] Within our model, there is no net charge on each dimer but there is a dipole associated with each dimer,

and consequently the overall structure has an electric polarization. One orientation of the dipole points along the protofilament axis, this is the up state (Fig. 2). We have some freedom to choose the direction of the dipole in the down state since the electric dipole has not been measured in each of tubulin's conformational states. We have chosen the down state's orientation in a direction which is roughly opposite to the direction of the up state but which is 29° from the vertical and points out radially away from the microtubule. This choice is based on the geometry of microtubule ends that are splaying apart during disassembly. Under these conditions, tubulin adopts a different conformation with curved protofilaments.^[17] The results of simulations presented here have been carried out with $\phi = 29^\circ$ down states. Some degree of tilting is necessary if action potentials are to influence the dipole dynamics because the non-tilted dipole is entirely axial and the electric field of the action potential is purely radial.^[18] Consequently, there would be no interaction unless one of the states has some radial component since the interaction energy between a dipole and an electric field is simply

$$U_{\text{int}} = -\bar{p} \cdot \bar{E}, \quad (1)$$

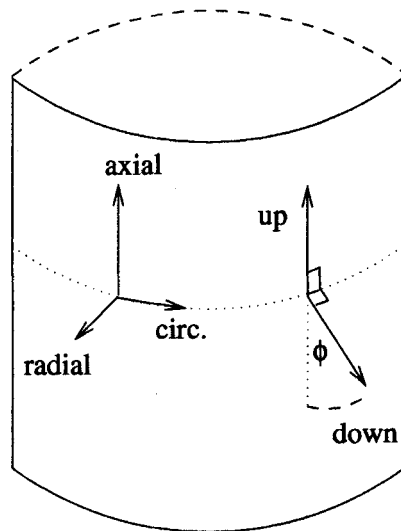


FIGURE 2 This view of the outer surface of a microtubule identifies the directions of fields we consider. The axial or vertical direction in this diagram points away from the cell body. At the right, the up and down directions which were selected for tubulin's dipole moment are shown ($\phi = 29^\circ$).

where $\bar{p}$ is the dipole moment and $\bar{E}$ is the electric field. The interaction energy between two elementary dipoles is given by the following well-known formula:^[19]

$$U_{\text{int}} = \frac{1}{4\pi\epsilon\epsilon_f} \frac{\bar{p}_1 \cdot \bar{p}_2 - 3(\bar{p}_1 \cdot \bar{n})(\bar{p}_2 \cdot \bar{n})}{r^3}, \quad (2)$$

where ϵ is the relative permittivity of the medium, ϵ_f is the permittivity of free space, $\bar{n}$ is the normal vector pointing from the position of the first dipole to the second dipole and r is the distance separating the dipoles. Consider the interaction between dipoles of the microtubule lattice: From a purely structural point of view, the lattice type may seem insignificant, as one does not expect that it will change assembly dynamics or elastic properties much. However, from the point of view of dipole interaction, the lattice type is crucial. In the B lattice, dimers are aligned in nearly tangential rows (Fig. 3). As a result, there is a strong interaction between neighboring protofilaments which favours antiparallel dipole orientations. In the A lattice, the neighboring protofilaments are shifted vertically such that identical orientation of the dipoles is favoured. In either case, there is a strong interaction along individual protofilaments which favours similarly oriented dipoles. The effect of the dipole interactions is that the lattice of up and down dipoles will self-organize into an energetically favourable configuration. At high temperatures, the lattice has a random state – each individual dimer can be found in the up or down state in a rather arbitrary fashion. At lower temperatures, the interaction energy dictates the dipole

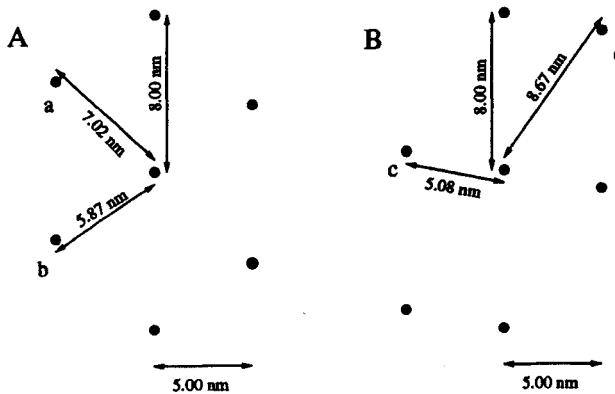


FIGURE 3 View of lattice structures. Note that the A lattice on the left is nearly perfectly triangular while the B lattice on the right is nearly rectangular. Shown as a planar lattice in the diagram, the lattice curls around in a cylindrical fashion to form the hollow microtubule.

arrangement and the lattice becomes ordered. Our task is to investigate this transition in order to determine the likely state of a microtubule under physiological conditions.

The lattice dynamics in our model are simulated using a standard Monte Carlo technique.^[20] For each timestep, the energy of the present state and the opposite state are calculated. A change of state is a random event; the probability of its occurrence is determined by the temperature. There are two numbers which are put into the model by hand. One is the relative permittivity of tubulin in cytoplasm, ϵ and the other is the magnitude of the dipole moment of tubulin. The dielectric constant of dry tubulin or of tubulin in solution has not been found in the literature although some experiments give permittivities in excess of 100 for some other proteins in solution.^[21] Our choice of these parameters allows us to compare results with previous modelling attempts. All of this dependence can be accounted for by a single parameter, Q , where

$$Q = p^2 / \epsilon . \quad (3)$$

The dynamics of the model are unaffected by the choice of Q but it simply scales with the temperature. Larger values of Q increase interaction strengths and thereby increase the transition temperature. We have chosen $Q = 12 \times 10^{-56} \text{C}^2 \cdot \text{m}^2 \simeq 10800 \text{ (Debye)}^2$. For a dipole corresponding to an electronic charge, with a charge separation of one dimer unit, 8 nm, $\epsilon \simeq 13$ given this choice. Such a value for the dipole moment has the same order of magnitude as some other globular proteins.^[21] This may be a conservative estimate as Athenstädt has reported the dipole moment of another fibrous molecule, collagen, to be about 15 000 Debye. The important number is not the total dipole but the change in the dipole, consequently the smaller figure that we report seems reasonable. The precise value of the dipole moment depends on the pH of the surrounding medium. The following results are derived from these estimates of the tubulin dimer's dipole moment. We shall comment on how any changes to these parameters affect the results of our simulations.

3. RESULTS

The results presented represent a summary of our work, and the most important features are highlighted. A more extensive look at the microtubule lattices and their ferroelectric ordering may be found in our

previous publications.^[22–24] We are plotting an order parameter, P , representing scaled polarization, which is defined over the N lattice sites by

$$P = \frac{1}{N} \sum_{i,j} \sigma_{ij}, \quad (4)$$

where the σ_{ij} is $+1$ or -1 and describes the conformation of the dipole labelled (i, j) of the 2D cylindrical lattice. The results for the 13A lattice and the 13B lattice (as illustrated in Fig. 1) are shown in Figures 4 and 5. In addition, the polarization *versus* temperature graph is displayed for the 13A lattice as a function of an applied external electric field in Figure 6. We have also computed polarization curves in the case of lattice defects, both impurities and microtubules with alternative protofilament numbers (12 and 14 being the most common of such defects).

Along a protofilament, head to tail alignment of dipoles is preferred, so that the positive end of one dipole is next to the negative end of the adjoining dipole. Consequently, dipoles will seek parallel alignment along a protofilament at low temperature when the microtubule lies in its ground state. In the microtubule lattice, each dimer has six nearest neighbours due to the triangular nature of the lattice. Consider the two nearest neighbouring dimers which are located on the protofilament directly to the left of the protofilament of the dimer, labelled a and b (Fig. 3). Relative to the

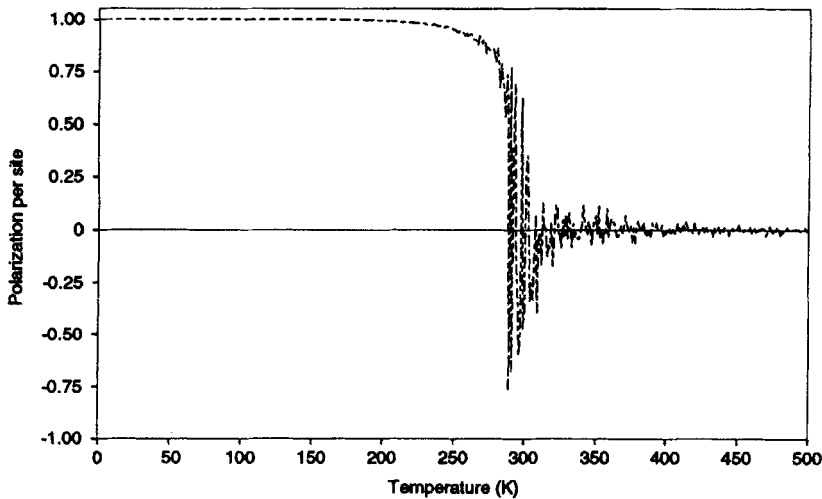


FIGURE 4 Relative electric polarization of the microtubule 13A lattice as a function of temperature, obtained from Monte Carlo modelling.

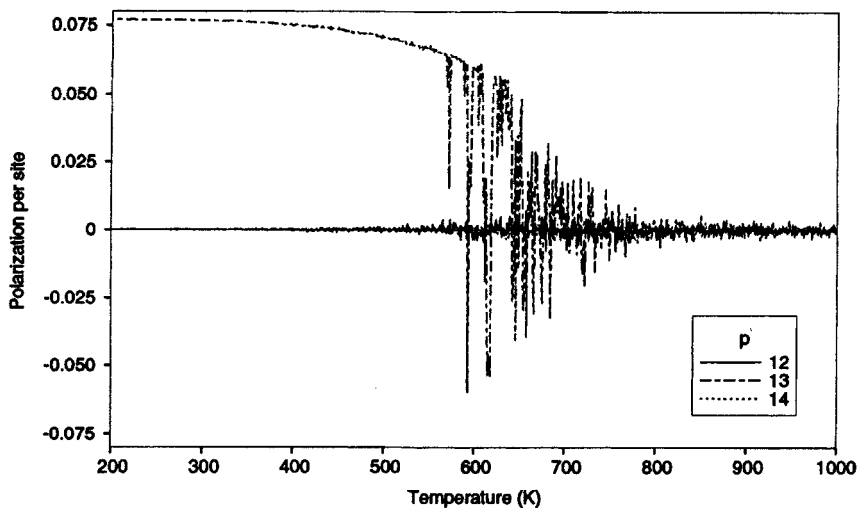


FIGURE 5 Polarization of the microtubule 12B, 13B and 14B lattices as a function of temperature. The 12B and 14B lattices show only random fluctuations about zero relative polarization because the ordered low temperature state consists of antiferroelectric protofilaments. In the 13B lattice, one protofilament of 13 must be unpaired, resulting in a finite polarization.

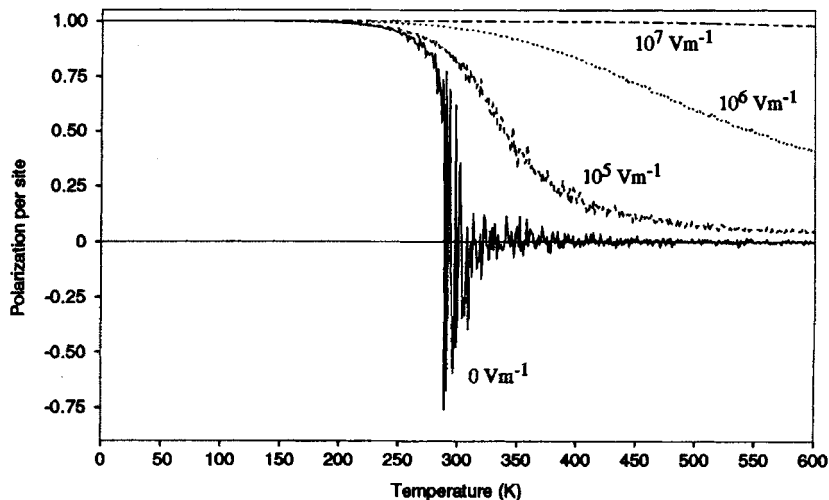


FIGURE 6 Polarization of the microtubule 13A lattice as a function of temperature as an electric field is applied in the direction of the up moment.

interaction along the protofilament, J , there is an interaction of strength $0.35J$ along the direction a , and an interaction of strength $-0.22J$ along the direction b . Both of these are smaller than the interaction along the

protofilament, so indeed parallel alignment along the protofilament is expected. Of the off-protofilament interactions, the larger interaction (a) is ferroelectric so neighbouring protofilaments adopt a configuration of parallel dipoles. Consequently, the ground state consists of all dipoles in parallel alignment (Fig. 8a). In the B lattice, the situation is changed; there is an interaction of strength -1.76 J along the direction c , and an interaction of strength 0.39 J along the direction d (Fig. 3). This direction, c , must have alternating dipoles. Given that dipoles alternate along the helix defined by the d direction, the other two interactions are conflicting but the interaction along the protofilament is larger. As a result, dipoles are still in parallel alignment along the protofilament. The entire lattice now consists of alternating protofilaments (Fig. 8b).

3.1. Lattice Defects

In addition to the effects of lattice type and applied fields upon the ordering of the lattice, there is also the question of how defects in the lattice have an impact upon dipole ordering. Firstly, consider lattice defects where the protofilament number is different than the usual 13. In the case of such a microtubule, there is now a seam for both the A and B lattices. In the 14A lattice, the interaction at the seam is now B-lattice-like. As a result, at the

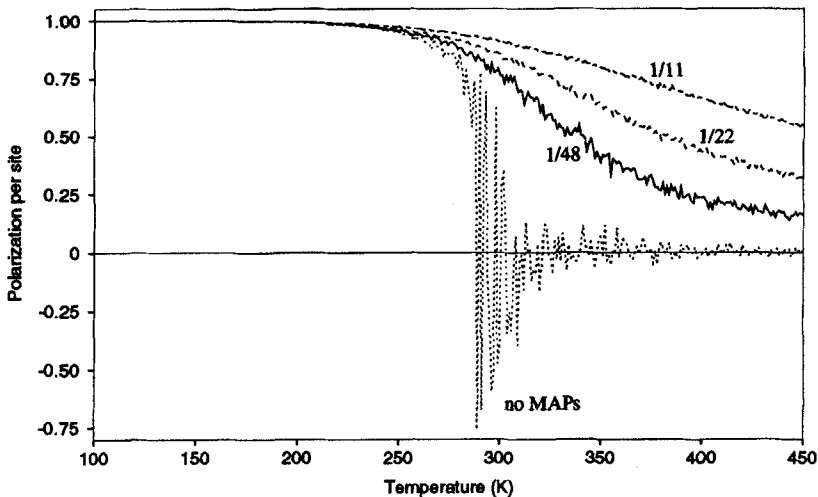


FIGURE 7 Polarization of the microtubule 13A lattice as a function of the ratio of microtubule-associated proteins to tubulin dimers. Ratios taken from Dustin^[25] and 1/22 means one in each 22 dimers is attached to a MAP and held in the up conformation.

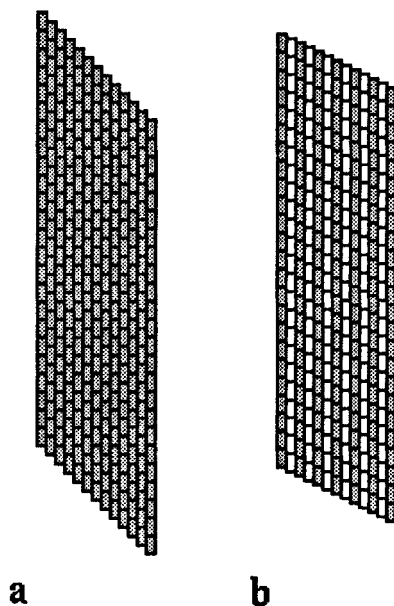


FIGURE 8 (a) Ground state of the microtubule 13A lattice, which is completely ferroelectric. Dimers are illustrated and dark boxes and light boxes indicate states of opposite dipole moments. (b) Ground state of the microtubule 13B lattice, which demonstrates antiferroelectric order between neighbouring protofilaments. Dimers are illustrated as in (a).

seam the ordering between the neighbouring protofilaments favours antiparallel ferroelectric order. These two protofilaments adopt the antiparallel ferroelectric ordering and the remaining protofilaments have some flexibility. That is, a ferroelectric seam must exist on the microtubule due to the dipole frustration. It should be distinguished from the seam we have discussed up until this point, which was structural; its location is immaterial since it does not have an energetic consequence. So, one drawback to such a lattice is the relative lack of order since dipole interactions alone do not make all microtubules with 14 protofilaments and an A lattice alike. Similarly, if we consider the 14B lattice, the bulk of the lattice imposes that neighbouring protofilaments should have alternating dipole orientations; but at the structural seam which closes the microtubule cylinder the preferred interaction is of antiparallel dipole orientations. The ground state occurs when the frustration is localized at the structural seam; however, the net polarization of the microtubule defined by Eq. (4) is zero since the numbers of dimers in the up state and in the down state are equal. Since the dipole of the up state and down state are not exactly opposite, the

microtubule does retain a small polarization. Thus the lattice type and the boundary conditions of the structural seam play an important role in determining the ordered state of the polymer.

We have also studied the impact which microtubule associated proteins (MAPs) have on dipole ordering in microtubules. MAPs attach themselves to microtubules in periodic patterns.^[25] Their function is to stabilize the assembly properties of the microtubule. Our hypothesis is that this is accomplished by preventing the conformational change of the tubulin dimer to which it is attached. The net effect of such MAPs has been simulated given this assumption, and the result is qualitatively similar to the application of an electric field (Fig. 7). This may be understood by viewing the fixed conformation dipoles as seeds of order. In the limit where one could connect a MAP to each tubulin dimer in a microtubule, the lattice would be ordered at all temperatures.

One thing which has not yet been experimentally established is the magnitude of the dipole moment on the tubulin dimer. It seems quite likely that it is sufficiently large for dipole interactions to cause ordering of the molecular conformations of the microtubule. Indeed, it is even possible that, due to the shielding of the microtubule by ordered water, the effective temperature experienced by the polymer would be somewhat lower and dipolar effects would be even more important.

Our investigations show that the microtubule is also sensitive to electric fields of the order 10^5 V/m; this may be more usefully understood as 100 mV/ μ m and is smaller than the typical electric field of 10^7 V/m measured across cellular membranes. If the MAPs function as we have hypothesized, then weaker fields would suffice. The significance of this observation comes mainly from the geometry of microtubules in nerve cells, which pass electrical signals (nerve impulses) along their membranes and give rise to transient electric fields. These short-lived fields may be sufficient to cause a wave of reordering along the microtubule, just as a wake follows a ship. It is essential that the microtubule has a highly ordered resting state if such signals are to be interpreted by receptor proteins at the synaptic junction between two nerve cells. That is to say, a signal could not be distinguished from the background unless the background was relatively smooth. This may indeed be the function of the dipole interactions.^[18] Our simulations show signal processing or transmission as possibilities but rule out the storage of information as Hameroff *et al.*^[2] have hypothesized. The main problem with information storage through particular patterns of dipole states is the need to shield the system from thermal fluctuations. In our model this means operation below the transition temperature. Below the

transition temperature, signals are only transient because the microtubule must ultimately reside in its ground state.

4. DISCUSSION

The ferroelectric properties of the microtubule seem to be important in determining the orientation of the microtubule, may be important in directing the traffic of motor proteins on their outer surface and are also essential if signals are passed along the microtubule by a wave of conformational change. While the electromagnetic hand may not be required for this direction, it does simplify the explanation of certain signalling properties such as a synchronizing signal prior to chromosome separation or signal propagation along an axonal microtubule. The phase transition from a disordered to an ordered lattice of dipoles is not necessarily observable within a living cell, since one expects that the microtubule would always remain ordered to retain its function. However, the transition temperature does determine the magnitude of the dipole required for electromagnetic interaction to be the dominant influence on lattice dipole dynamics. The larger the dipole, the more important both the dipole-dipole interaction and the interaction with external electric fields.

Currently investigation is underway to examine the conduction properties of the microtubule. Should the microtubule be found to have semiconductor properties, this electron mobility will lend much weight to the ferroelectric explanation of microtubule function.

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